

AMENDMENT NO. 1 AUGUST 2002

TO

IS 12438 : 1988 SPECIFICATION FOR DIMETHYL TEREPHTHALATE

(*Page 1, clause 1.1*) — Insert the following new clause after 1.1 and renumber the subsequent clauses:

‘2 REFERENCES

The following standards contain provision which through reference in the text, constitute provisions of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

<i>IS No.</i>	<i>Title</i>
2362 : 1993	Determination of water by the Karl Fischer method — Test method (<i>first revision</i>)
13320 : 1992	Determination of iron by atomic absorption spectrophotometry — Test method’

[*Page 1, Table 1, Sl No. (iii)*] — Substitute the following for the existing:

‘(1)	(2)	(3)	(4)
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iii) Acid number (mg, KOH/g)	0.03	A-5’	
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[*Page 1, Table 1, Sl No. (iv)*] — Substitute the following for the existing:

‘(1)	(2)	(3)	(4)
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iv) Moisture content, percent by mass, <i>Max</i>	0.02	IS 2362 : 1993’	
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(*Page 4, clause A-5.5*) — Substitute the following for the existing formula:

$$\text{Acid number (mg, KOH/g)} = \frac{V \times N \times 56.1}{M}$$

(*Page 4, clause A-6*) — Delete.

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Amend No. 1 to IS 12438 : 1988

(Page 4, clause A-7) — Substitute the following for the existing clause:

A-7 DETERMINATION OF IRON CONTENT

A-7.0 General

Two methods have been specified for determination of iron content, namely:

- a) Method 1 — Photometric method; and
- b) Method 2 — Atomic absorption spectrophotometric method.

Both the methods can be used for determination of iron on routine basis. However in the event of any dispute, Method 2 should be treated as a referee method.

A-7.1 Method 1 : Photometric Method

A-7.1.1 Outline of the Method

Pure dimethyl terephthalate is incinerated and the iron contained in the ash is converted into iron thioglycollate. The concentration is determined photometrically.

A-7.1.2 Apparatus

A-7.1.2.1 Spectrophotometer

A-7.1.3 Reagents

A-7.1.3.1 Thioglycollic acid solution — 10 percent. If after its preparation, the solution becomes turbid, it should be filtered.

A-7.1.3.2 Sodium citrate solution — Dissolve 150 g of AR grade trisodium citrate in water and make up to 500 ml. Transfer this solution into a 1 000-ml separating funnel, add 0.5 ml of concentrated ammonium hydroxide solution and add 50 ml of 0.02 percent dithiozone in chloroform solution, and shake well. After phase separation, extract the aqueous phase several times, using 50 ml of diluted dithiozone solution (10 ml of 0.02 percent dithiozone solution and 40 ml of chloroform for each extraction). Then add 5 ml of 20 percent citric acid solution and shake with 50 ml of chloroform each time until the final extract is colourless. Store the solution in a polyethylene bottle.

Amend No. 1 to IS 12438 : 1988

A-7.1.3.3 Standard iron solution — Dissolve 0.7023 g of pure ammonium ferrous sulphate [$\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$] in water contained in a 1 000-ml measuring flask. Add 5 ml of sulphuric acid and dilute the contents to 1 000-ml.

Further dilute 25 ml of this solution to 250 ml. 1 ml of prepared solution corresponds to 10 μg of iron.

A-7.1.3.4 Preparation of calibration chart — Fill a set of 50-ml measuring flasks with 2.0 ml, 4.0 ml, 8.0 ml, 12.0 ml, 16.0 ml and 20.0 ml of standard iron solution and add 5 ml of dilute sulphuric acid into each flask.

Further add 2 ml of sodium citrate solution and dilute the contents with water to 25 to 27 ml. Add 5 ml of concentrated hydrochloric acid and while cooling the solution, add concentrated ammonium hydroxide solution drop-by-drop until a slightly alkaline reaction is obtained. Now add another 1.5 ml of hydrochloric acid in excess.

Cool the contents well and mix 1.0 ml of 10 percent thioglycollic acid solution, then add 3.0 ml of concentrated ammonium hydroxide solution and make up the solution to 50 ml.

The reference solution is prepared in the same manner but without the iron solution.

After the solutions have been allowed to stand for 5 minutes, measure the absorption of each solution in a 4-cm cell at a wavelength of 535 nm, using the iron-free solution for reference. On the basis of the values obtained, prepare a chart by plotting the iron content in μg as a function of the measured absorption.

A-7.1.3.5 Procedure — Incinerate 30 g of the material in a quartz crucible. After the flame is extinguished, transfer the crucible to a muffle furnace maintained at 550°C and continue incineration for 30 minutes. Allow it to cool and dissolve the contents in 10 ml of hydrochloric acid solution (1:1) by adding through the sides of the dish. Bring it to boil on a hot plate. Transfer it to 50-ml measuring flask. Wash with distilled water three times and transfer to the measuring flask. Add 2 ml of sodium citrate solution followed by 1 ml of 10-percent thioglycollic acid solution. Add excess of ammonia solution to ensure that the solution remained alkaline. Make up to the mark with distilled water and mix thoroughly. At the end of 5 minutes, measure the absorption at 535 nm using the spectrophotometer.

Amend No. 1 to IS 12438 : 1988

A-7.1.4 Calculations

$$\text{Iron content, parts per million} = \frac{A}{M}$$

where

A = iron content in μg as read from the chart, and

M = mass of the sample taken for the test.

NOTES

1 If the iron concentration is low, extreme care is taken to prevent foreign matter from getting into the sample during the incineration.

2 If the concentration is so high that on comparison with the 4-cm cell the linear range is exceeded, measurements should be repeated using a diluted solution.

A-7.2 Method 2 : Atomic Absorption Spectrophotometric Method

A-7.2.1 Outline of the Method

About 100 g of DMT in powdered form is sublimed in platinum dish using hot plate. A platinum dish with unsublimed content is incinerated in muffle furnace. Ash obtained is dissolved in dilute hydrochloric acid and made up to suitable volume. The solution thus obtained is used for determination of iron using atomic absorption spectrophotometer (*see* also IS 13320 : 1992).

A-7.2.2 Apparatus

A-7.2.2.1 Platinum dish — 50-ml capacity.

A-7.2.2.2 Muffle furnace

A-7.2.2.3 Hot plate

A-7.2.2.4 Micro balance

A-7.2.2.5 Atomic absorption spectrophotometer

A-7.2.3 Reagents

A-7.2.3.1 Hydrochloric acid solution — (1:1 V).

Amend No. 1 to IS 12438 : 1988

A-7.2.3.2 Iron Standard 3 ppm — Dissolve 0.0211 g of pure ammonium ferrous sulphate $[\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}]$ in distilled water. Add 5 ml of 1:1 HCl and dilute to 1 000 ml with distilled water.

A-7.2.4 Procedure — Weigh about 100 g of powdered DMT (S g) and put small quantity of it in clean platinum dish. Place platinum dish on hot plate and allow DMT to sublime in a fume cupboard. Repeat with remaining quantity of DMT till the entire weighed sample is consumed. Transfer platinum dish to muffle furnace at 550°C for 15 minutes. Remove the dish from furnace and allow it to cool in a desiccator. Dissolve the contents in 10 ml (1:1) hydrochloric acid and warm it using hot plate. Transfer it to 25-ml volumetric flask. Make up the volume by rinsing platinum dish with distilled water. Prepare a blank in 25-ml volumetric flask by taking 10 ml hydrochloric acid (1:1) and make up the volume with distilled water. Analyze iron using atomic absorption spectrophotometer.

A-7.2.5 Calculation

$$\text{Iron, ppm} = \frac{\text{Absorbance of sample}}{\text{Absorbance of standard}} \times 3 \times \frac{25}{S}$$

where

S = mass of DMT sample in g.

(Page 5, clause A-8.2, line 13) — Substitute 'to the nearest 0.1 mg' for 'to nearest 0.1 g'.